

Derivative of Transform

Let

$$\boxed{L[f(t)] = F(s)} \quad - (1)$$

then,

$$\boxed{L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} [L\{f(t)\}]} \quad L(*)$$

$$n = 1, 2, 3, \dots$$

Ex #01

$$L[t e^{3t}] = ?? \quad L(1)$$

Method #01

$$L[t e^{3t}] = \int_0^{\infty} t e^{3t} e^{-st} dt$$

Sol Method #02 (above formula)

$$L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} [L(f(t))] \quad L(2)$$

Let's $\nearrow$ compute (1) & (2),

$$f(t) = e^{3t} \quad - (3)$$

$$L[f(t)] = L[e^{3t}] \quad - (4)$$

$$F(s) = \frac{1}{s-3} \quad - (5)$$

put (5) in (2),

$$(2) \Rightarrow L[t f(t)] = L[t e^{3t}] = (-1)^1 \frac{d}{ds} \left[\frac{1}{s-3} \right] \quad L(6)$$

$$\Rightarrow L[t e^{3t}] = (-1) (s-3)^{-1-1} (-1)$$

$$\boxed{L[t e^{3t}] = \frac{1}{(s-3)^2}} \quad (*)$$

Ex $L[t^2 \sin kt] = ? \quad \text{--- (1)}$

Sol $\boxed{n=2}$ Now compute LHS of (1) & (2)

$$\Rightarrow L[t^2 f(t)] = (-1)^2 \frac{d^2}{ds^2} [L(f(t))] \quad L(2)$$

(2) $f(t) = \sin kt \quad \text{--- (3)}$

$$F(s) = L[f(t)] = \frac{k}{s^2 + k^2} \quad \text{--- (4)}$$

put (4), (3) in (2)

$$(2) \Rightarrow L[t^2 \sin kt] = (-1)^2 \cdot \frac{d^2}{ds^2} \left[\frac{k}{s^2 + k^2} \right]$$

$$= \frac{-2k^3 + 6ks^2}{(s^2 + k^2)^3} \quad (\text{Exercise})$$

Practice problems

Find Following

Q1 $L[t \cos wt]$, Q2 $L[3 \sinh 4t]$, Q3 $L[t e^t \cos t]$

(6.2 § 8.12)
Application of Laplace Transform of Integral

Let if $\boxed{L[f(t)] = F(s)}$, then
 L^{-1}

$$\Rightarrow \boxed{L\left[\int_0^t f(\eta) d\eta\right] = \frac{F(s)}{s}} \quad (2)$$

then, $\boxed{L^{-1}\left[\frac{F(s)}{s}\right] = \int_0^t f(\eta) d\eta} \quad (3)$

where " η " is variable of integration.

Ex #01

Find Laplace inverse transform of

$$\boxed{F(s) = \frac{1}{s(s^2 + w^2)}} \quad (1)$$

Sol

Method #01

$$\boxed{\frac{1}{s(s^2 + w^2)} = \frac{A}{s} + \frac{Bs + C}{s^2 + w^2}} \quad (2)$$

Method #02 use above integral formula.

Recall

$$\mathcal{L}[f(t)] = F(s) \quad (1)$$

$$f(t) = \mathcal{L}^{-1}[F(s)] \quad (2)$$

$$\int_0^t f(\eta) d\eta = \mathcal{L}^{-1}\left[\frac{1}{s} \cdot F(s)\right] \quad (3)$$

(F(s) gives us information about f(η))

(Just replace t by η and integrate)

Recall
⇒ Question

$$\int_0^t (??) d\eta = \mathcal{L}^{-1}\left[\frac{1}{s} \cdot \frac{1}{s^2 + \omega^2}\right] \quad (4)$$

by comparing (3) & (4)

$$F(s) = \frac{1}{s^2 + \omega^2} \quad (5)$$

$$\mathcal{L}^{-1}[F(s)] = \frac{1}{\omega} \mathcal{L}^{-1}\left[\frac{\omega}{s^2 + \omega^2}\right] \quad (6)$$

$$\Rightarrow \boxed{f(t) = \frac{\sin \omega t}{\omega}}$$

$\mathcal{L}^{-1}$ (7)

$$\boxed{f(\eta) = \frac{\sin \omega \eta}{\omega}}$$

$\mathcal{L}^{-1}$ (8)

put (8) on L.H.S of (4)

(4) $\Rightarrow$

$$\int_0^t \frac{\sin \omega \eta}{\omega} d\eta = \mathcal{L}^{-1} \left[\frac{1}{s(s^2 + \omega^2)} \right] \quad \mathcal{L} (9)$$

$$\mathcal{L}^{-1} \left[\frac{1}{s(s^2 + \omega^2)} \right] = -\frac{1}{\omega} \left[\frac{\cos \omega \eta}{\omega} \right]_{0=\eta}^{t=\eta}$$

$$= -\frac{1}{\omega^2} (\cos \omega \eta - 1)$$

$$\mathcal{L}^{-1} \left[\frac{1}{s(s^2 + \omega^2)} \right] = \frac{1 - \cos \omega \eta}{\omega^2} \quad (10)$$

Example #02

$$\mathcal{L}^{-1} \left[\frac{1}{s^2(s^2 + \omega^2)} \right] = ?? \quad (11)$$

Sol

01

$$\mathcal{L}^{-1} \left[\frac{1}{s(s^2 + \omega^2)} \right] = \mathcal{L}^{-1} \left[\frac{1}{s} \cdot \frac{1}{s^2 + \omega^2} \right] = \frac{1 - \cos \omega \eta}{\omega^2} \quad (12)$$

02

$$\mathcal{L}^{-1} \left[\frac{1}{s^2(s^2 + \omega^2)} \right] = \mathcal{L}^{-1} \left[\frac{1}{s} \cdot \left(\frac{1}{s(s^2 + \omega^2)} \right) \right] \quad (13)$$

$$= \int_0^t \frac{1 - \cos \omega \eta}{\omega^2} d\eta \quad (14)$$

$$= \frac{1}{\omega^2} \left[\int_0^t d\eta - \int_0^t \frac{\cos \omega \eta}{\omega} d\eta \right]$$

$$= \frac{1}{\omega^2} \left[t - \frac{1}{\omega} \left(\sin \omega \eta \Big|_0^t \right) \right]$$

$$= \frac{1}{\omega^2} \left[t - \frac{1}{\omega} (\sin \omega t - 0) \right]$$

$$\boxed{\mathcal{L}^{-1} \left[\frac{1}{s^2(s^2 + \omega^2)} \right] = \frac{t}{\omega^2} \left[t - \frac{\sin \omega t}{\omega} \right]}$$

Ex = $\mathcal{L}^{-1} \left[\frac{1}{s(s^2 + 4)} \right] \quad \text{--- (1)}$

Sol $\mathcal{L}^{-1} \left[\frac{1}{s} \cdot \frac{1}{s^2 + 4} \right] = \int_0^t \frac{\sin 2\eta}{2} d\eta \quad \text{--- (2)}$

$$\boxed{\mathcal{L}^{-1} \left[\frac{1}{s} \cdot F(s) \right] = \int_0^t f(\eta) d\eta} \quad \text{--- (*)}$$

$$F(s) = \frac{1}{s^2 + 4} \Rightarrow \mathcal{L}^{-1} [F(s)] = \frac{1}{2} \mathcal{L}^{-1} \left[\frac{2}{s^2 + 2^2} \right] \quad \text{--- (3)}$$

$$\boxed{f(t) = \frac{\sin 2t}{2}} \quad \text{--- (5)}$$

$$(2 \Rightarrow) \mathcal{L}^{-1} \left[\frac{1}{s(s^2+4)} \right] = \frac{1}{2} \left. -\frac{\cos 2t}{2} \right|_0^t \quad (6)$$

$$= -\frac{1}{4} (\cos 2t - 1) \quad (7)$$

$$= \frac{1 - \cos 2t}{4} \quad (8)$$

Exercise

$$\underline{\underline{Q1}} \quad \mathcal{L}^{-1} \left[\frac{1}{s^2(s+9)} \right]$$

$$\underline{\underline{Q2}} \quad \mathcal{L}^{-1} \left[\frac{1}{s^2(s^2-1)} \right]$$

Laplace transform of Derivative

Let

$$L[f(t)] = F(s) \quad (1)$$

$$\text{or } \boxed{L[y(t)] = Y(s)} \quad (2)$$

then,

$$\boxed{L[y'(t)] = sY(s) - y(0)} \quad (3)$$

$$\boxed{L[y''(t)] = s^2 Y(s) - sy(0) - y'(0)} \quad (4)$$

$$\boxed{L[y'''(t)] = s^3 Y(s) - s^2 y(0) - sy'(0) - y''(0)} \quad (5)$$

Ex #01 Solve

$$y'' + 4y = 16t \quad \text{--- (1)}$$

Solution

$$y(0) = 0, \quad y'(0) = -6 \quad \text{--- (2)}$$

We take Laplace transforms of both sides of the given differential equation. Thus we obtain,

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y\} = \mathcal{L}\{16t\} \quad \text{--- (2)}$$

We use the formulae: $\mathcal{L}\{y''\} = s^2 Y(s) - sy(0) - y'(0)$, $\mathcal{L}\{y\} = Y(s)$, $\mathcal{L}\{t\} = \frac{1}{s^2}$.

Therefore, the above equation becomes,

$$[s^2 Y(s) - sy(0) - y'(0)] + 4Y(s) = 16(1/s^2) \quad \text{--- (6)}$$

Thus using the given initial conditions we obtain,

$$[s^2 Y(s) - 3s + 6] + 4Y(s) = 16/s^2 \quad \text{--- (7)}$$

i.e.,

$$(s^2 + 4)Y(s) = 3s - 6 + 16/s^2 \quad \text{--- (8)}$$

or,

$$Y(s) = \frac{3s-6}{s^2+4} + \frac{16}{s^2(s^2+4)} \quad \text{--- (9)}$$

$$Y(s) = \frac{3s}{s^2+4} - \frac{6}{s^2+4} + \frac{16}{4} \left(\frac{1}{s^2} - \frac{1}{s^2+4} \right) \quad \text{--- (10)}$$

$$\therefore \frac{16}{s^2(s^2+4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+4}$$

$$Y(s) = \frac{3s}{s^2+4} - \frac{10}{s^2+4} + \frac{4}{s^2} \quad \text{--- (11)}$$

Therefore, by taking the inverse Laplace transform of both sides we obtain,

$$\mathcal{L}^{-1}[Y(s)] = 3\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} - 5\mathcal{L}^{-1}\left\{\frac{2}{s^2+4}\right\} + 4\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} \quad \text{--- (12)}$$

But we know that,

$$\mathcal{L}\{\cos 2t\} = s/(s^2+4), \quad \mathcal{L}\{\sin 2t\} = 2/(s^2+4), \quad \mathcal{L}\{t\} = 1/s^2.$$

Therefore, the solution of the given differential equation is,

$$y(t) = 3 \cos 2t - 5 \sin 2t + 4t \quad \text{--- (16)}$$

Example 8.1.2

Ex #02

Solve, $y'' + y = 16 \cos t$, $y(0) = 0$, $y'(0) = 0$.

$\underbrace{\hspace{1cm}}_{(1)} \quad \underbrace{\hspace{1cm}}_{(2)}$

Solution

Taking Laplace transform of both sides the given equation can be written as,

$$\mathcal{L}\{y''\} + \mathcal{L}\{y\} = 16 \mathcal{L}\{\cos t\} \quad \text{--- (3)}$$

or,

$$[s^2 Y(s) - s y(0) - y'(0)] + Y(s) = 16 s / (s^2 + 1). \quad \text{--- (4)}$$

By using the initial conditions it becomes,

$$s^2 Y(s) + Y(s) = 16s / (s^2 + 1) \quad \text{--- (5)}$$

i.e.,

$$Y(s) (s^2 + 1) = 16s / (s^2 + 1). \quad \text{--- (6)}$$

Thus,

$$Y(s) = \mathcal{L}\{y(t)\} = \frac{16s}{(s^2 + 1)^2} \quad \text{--- (7)}$$

Taking inverse Laplace transform of both sides we get,

$$y(t) = 8 \mathcal{L}^{-1} \left\{ \frac{2s}{(s^2 + 1)^2} \right\}. \quad \text{--- (8)}$$

$$\therefore \mathcal{L}[\sin kt] = \frac{k}{s^2 + k^2}$$

But we know that, $\mathcal{L}\{t \sin kt\} = \frac{2ks}{(s^2 + k^2)^2}$, $s > 0$. Thus we get the solution as,

$$y(t) = 8 t \sin t. \quad \text{--- (9)}$$

$\underbrace{\hspace{1cm}}_{(10)}$

Example 8.10.3

Solve, $y'' + 2y' + 5y = 0$, $y(0) = 3$, $y'(0) = -7$.

$\underbrace{\hspace{1cm}}_{(1)} \quad \underbrace{\hspace{1cm}}_{(2)} \quad \underbrace{\hspace{1cm}}_{(3)}$

Solution

Taking Laplace transform of both sides of the given equation we get,

$$[s^2 Y(s) - s y(0) - y'(0)] + 2 [s Y(s) - y(0)] + 5 Y(s) = 0. \quad \text{--- (4)}$$

Using the initial conditions this equation becomes,

$$[s^2 Y(s) - 3s + 7] + 2 [s Y(s) - 6] + 5 Y(s) = 0$$

$\underbrace{\hspace{1cm}}_{(5)}$

i.e.,

$$Y(s)(s^2 + 2s + 5) = 3s - 1. \quad \text{--- (6)}$$

Thus,

$$Y(s) = \frac{3s-1}{s^2+2s+5} = \frac{3(s+1)-4}{(s+1)^2+4} = 3\left[\frac{(s+1)}{(s+1)^2+4}\right] - 2\left[\frac{2}{(s+1)^2+4}\right]. \quad \text{--- (7)}$$

But we know that,

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+4}\right\} = \cos 2t, \quad \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+4}\right\} = e^{-t}\cos 2t, \quad \text{--- (8)}$$

$$\mathcal{L}^{-1}\left\{\frac{2}{s^2+4}\right\} = \sin 2t, \quad \mathcal{L}^{-1}\left\{\frac{2}{(s+1)^2+4}\right\} = e^{-t}\sin 2t. \quad \text{--- (9)}$$

Therefore, taking the inverse Laplace transform of the above equation we obtain the solution of the given differential equation as,

$$y(t) = e^{-t} [3 \cos 2t - 2 \sin 2t]. \quad \text{--- (12)}$$

Example 8.12.4

$$\text{Solve } y'' - 3y' + 2y = 12e^{4t}, \quad y(0) = 1, \quad y'(0) = 0. \quad \text{--- (1)}$$

Solution

Taking Laplace transform of both sides we can write the given equation as,

$$[s^2 Y(s) - s y(0) - y'(0)] - 3[s Y(s) - y(0)] + 2Y(s) = \frac{12}{s-4}. \quad \text{--- (2)}$$

Using the initial conditions it becomes,

$$[s^2 Y(s) - s] - 3[s Y(s) - 1] + 2Y(s) = \frac{12}{s-4}, \quad \text{--- (3)}$$

i.e.,

$$Y(s)(s^2 - 3s + 2) = s - 3 + \frac{12}{s-4}. \quad \text{--- (4)}$$

Thus,

$$Y(s) = \frac{s^2 - 7s + 24}{(s-1)(s-2)(s-4)} \quad \text{--- (6)}$$

Now, we resolve the term on the right into partial fractions. We let,

$$\frac{s^2 - 7s + 24}{(s-1)(s-2)(s-4)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-4}, \quad \text{--- (7)}$$

or,

$$s^2 - 7s + 24 = A(s-2)(s-4) + B(s-1)(s-4) + C(s-1)(s-2). \quad \text{--- (8)}$$

To find A we put $s-1 = 0$, i.e., $s = 1$ in the above equation. Thus we get, $18 = A(3)$, i.e.,

$$A = 6.$$

Similarly, to find B and C, we put $s = 2$, and $s = 4$ respectively in the above equation.

Accordingly we obtain, $B = -7$, and $C = 2$. Thus we get,

$$Y(s) = \frac{6}{s-1} - \frac{7}{s-2} + \frac{2}{s-4}. \quad \text{--- (9)}$$

Taking the inverse Laplace transform of both sides we get,

$$y(t) = 6 \mathcal{L}^{-1}\left(\frac{1}{s-1}\right) - 7 \mathcal{L}^{-1}\left(\frac{1}{s-2}\right) + 2 \mathcal{L}^{-1}\left(\frac{1}{s-4}\right) \quad \text{--- (10)}$$

Therefore,

$$y(t) = 6e^t - 7e^{2t} + 2e^{4t}. \quad \text{--- (11)}$$

Example 8.5

Solve $y'' + 2y' + y = 1$, $y(0) = 2$, $y'(0) = -2$

$$\mathcal{L}(1) \quad \mathcal{L}(2) \quad \mathcal{L}(-2)$$

Solution

Taking Laplace transform of both sides of the given equation we obtain,

$$[s^2 Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] + Y(s) = 1/s.$$

$$\mathcal{L}(3)$$

Using the initial conditions this equation becomes,

$$[s^2 Y(s) - 2s + 2] + 2[s Y(s) - 2] + Y(s) = 1/s;$$

or,

$$Y(s)(s^2 + 2s + 1) = 1/s + 2s + 2 = \frac{2s^2 + 2s + 1}{s};$$

or,

$$Y(s) = \frac{2s^2 + 2s + 1}{s(s^2 + 2s + 1)} = \frac{2s^2 + 2s + 1}{s(s+1)^2}.$$

$$\therefore \frac{2s^2 + 2s + 1}{s(s+1)^2} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

Resolving the right hand side expression into partial fractions we obtain,

$$Y(s) = \frac{1}{s} + \frac{1}{s+1} - \frac{1}{(s+1)^2}.$$

Taking the inverse Laplace transforms of both sides we get the solution of the given differential equation as,

$$y(t) = 1 + e^{-t} - t e^{-t}.$$

EXERCISE – 8.4

[Solution of Differential Equations by Laplace Transforms]

Note: Answers are given against the respective questions within square brackets.

Solve the following differential equations,

1. ~~$y''(t) - k^2 y(t) = 0, (k \neq 0)$~~

~~$[y = c_1 e^{kt} + c_2 e^{-kt}]$~~

2. ~~$y''(t) + k^2 y(t) = a, (k \neq 0);$~~

~~$[y = A \cos kt + B \sin kt + a/k^2]$~~

3. ~~$y''(t) + 2ky'(t) + k^2 y(t) = 0;$~~

~~$[y = e^{-kt} (c_1 + c_2 t)]$~~

Solve the following initial value problems,

4. $y'(t) - y(t) = 0, y(0) = -1;$

$[y = -e^t]$

5. $y'(t) - y(t) = e^t, y(0) = 1;$

$[y = (t+1)e^t]$

6. $y'(t) + y(t) = e^{-t}, y(0) = 1;$

$[y = (t+1)e^{-t}]$

7. ~~$y''(t) + 2ay'(t) + (a^2 + b^2)y(t) = 0, y(0) = 0, y'(0) = 1;$~~

~~$[y = \frac{1}{b} e^{at} \sin bt]$~~

8. $y''(t) + 4y(t) = \sin t, y(0) = 0, y'(0) = 0;$

$[y = \frac{1}{3} \sin t - \frac{1}{6} \sin 2t]$

9. $y''(t) + 4y'(t) + 4y(t) = 0, y(0) = 1, y'(0) = 1;$

$[y = (1+3t)e^{-2t}]$

10. $y''(t) - 2y'(t) + 5y(t) = 0, y(0) = 2, y'(0) = 4;$

$[y = (2 \cos 2t + \sin 2t)e^t]$

11. $y''(t) + y'(t) = t^2 + 2t, y(0) = 4, y'(0) = -2;$

$[y = \frac{t^3}{3} + 2e^{-t} + 2]$

12. $y''(t) - 2y'(t) + y(t) = 0, y(0) = 0, y'(0) = 2;$

$[y = 2te^t]$

13. $y''(t) + y'(t) + y(t) = t^2, y(0) = 1, y'(0) = 1;$

$[y = e^{-t/2} (\cos \frac{\sqrt{3}t}{2} + \frac{7\sqrt{3}}{3} \sin \frac{\sqrt{3}t}{3}) + t^2 - 2t]$

14. $y''(t) - 5y'(t) - 6y(t) = e^{3t}, y(0) = 2, y'(0) = 1; [y = \frac{10}{21}e^{6t} + \frac{45}{28}e^{-t} - \frac{1}{12}e^{3t}]$